Example 2:

The following staff readings were observed successively with a level. The instrument having been moved after the second, fifth and eighth readings.

0.675, 1.230, 0.750, 2.565, 2.225, 1.935, 1.835, 3.220, 3.115 and 2.875. The RL of BM is 100.000 m

Obtain the reduced levels of each point and apply usual check.

Brief explanation for tabulating the readings.

1. The readings are tabulated in their respective method of tabulation.
2. In every setting up of instrument the first reading taken as BS. Last reading taken as FS, and other between readings as IS
3. The instrument is shifted (or) moved after the 2nd , 5th , 8th readings. They are the F. S readings.
4. Therefore 1,3,6.9 are BS readings and the others are IS reading (4,7)
5. Final reading is a FS reading. (10)
6. In two settings with a change point, first set's last reading being FS and second set's first reading being BS. They are entered in a same line for continuity of leveling process.

Solution

(i) Height of collimation method

Station	BS	IS	FS	HC or HI	RL	Remarks
A	0.675			100.675	100.000	BM
B	0.750		1.230	100.195	99.445	CP_1
C		2.565			97.630	
D	1.935		2.225	99.905	97.970	CP_2
E		1.835			98.070	
F	3.115		3.220	99.880	96.685	CP_3
G			2.875		96.925	
Total	6.475		9.550			

Calculations:

HC for I set up = RL of A + BS of A 100.000 + 0.675 = 100.675

RL of B = HC for I set up – FS of B = 99.445

HC for II set up = RL of B + BS of B = 100.195

RL of C = HC for II set up – FS of C = 97.630

RL of D = HC for II set up - FS of D = 97.970

HC for III set up = RL of D + BS of D = 99.905

RL of E = HC for III set up – FS of E = 98.070

RL of F = HC for III set up – FS of F = 96.685

HC for IV set up = RL of F + BS of F = 99.880

RL of G = HC for IV set up – FS of G = 96.925

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